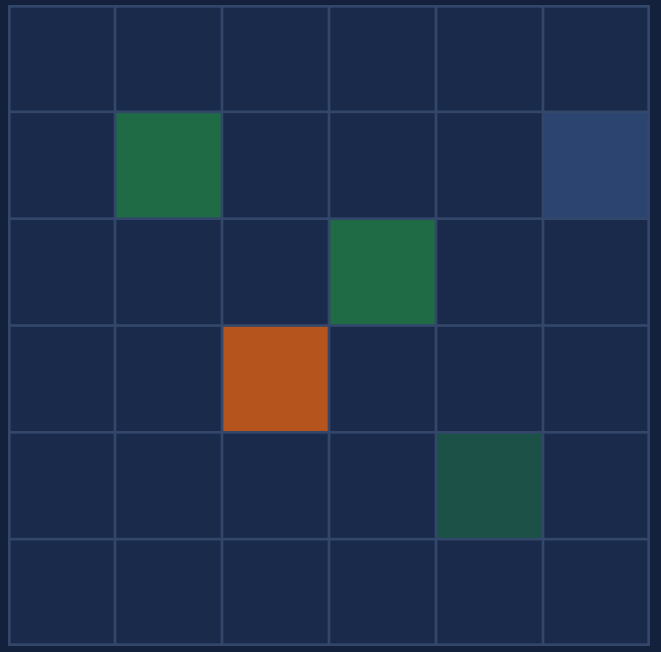
INF131 · TECHNOLOGY AND AI · WEEK 4

# Data quality and cleaning

Rows, columns, and how to turn messy data into evidence you can trust.



WHERE WE ARE

# From thinking in steps to clean data

WEEK 3

### Computational thinking

Break a problem into steps, conditions and repetition.

TODAY

### Data quality

Cleaning is the same idea: a fixed sequence of steps applied to every row.

LAB

### Power Query

Records each step, so the whole recipe can run again on new data.

A clean table is the input to everything that follows: pivot tables, SQL, Power BI and AI.

# What we mean by data

Data is a record of something that happened, written down so a computer can count it.

Each event below becomes **one row** in a table.

Each detail becomes **one column**.

### A sale in a shop

date · branch · product · units · price

### A country in a year

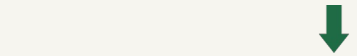
country · year · GDP per capita · inflation

### A bank loan

customer ID · age · income · amount · term

# Anatomy of an Excel sheet

Column C



| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Date | Branch | Product | Units | Price (ALL) |
| 2 | 03/10/2026 | Tirana | Coffee beans | 12 | 1,200 |
| 3 | 03/10/2026 | Durrës | Green tea | 5 | 650 |
| 4 | 04/10/2026 | Shkodër | Coffee beans | 8 | 1,200 |
| 5 | 04/10/2026 | Tirana | Olive oil | 3 | 1,500 |
| 6 | 05/10/2026 | Vlorë | Green tea | 10 | 650 |
| 7 | 05/10/2026 | Durrës | Olive oil | 6 | 1,500 |

### Row 4 = one record

One sale: 8 bags of coffee beans in Shkodër on 4 October.

### Column C = one variable

The same kind of fact for every row: which product was sold.

### Cell C4 = one value

Address = column letter + row number. Row 1 holds names, not data.

# One idea, three vocabularies

| Excel | Databases | Statistics and economics |
|---|---|---|
| Row | Record | Observation |
| Column | Field | Variable/Feature |
| Cell | Value | Data point |
| Sheet or table | Table | Dataset |

Rule of thumb: every row in a table describes the same kind of thing — one sale, one country-year, one customer.

PART 1 · DATA AND TABLES

# Three rules of tidy data

1

**Each variable is a column**

2

**Each observation is a row**

3

**Each value is a cell**

WHAT BREAKS THE RULES

Merged cells · subtotal rows in the middle of the data · two facts in one cell (“Tirana – 12 units”) · notes typed into data cells · colour used as information

# Same data, two shapes: wide and long

WIDE · one column per month

| Branch | Oct | Nov | Dec |
|---|---|---|---|
| Tirana | 120 | 135 | 160 |
| Durrës | 80 | 90 | 110 |



Unpivot

LONG · one row per branch-month

| Branch | Month | Units |
|---|---|---|
| Tirana | Oct | 120 |
| Tirana | Nov | 135 |
| Tirana | Dec | 160 |
| Durrës | Oct | 80 |
| Durrës | Nov | 90 |
| Durrës | Dec | 110 |

World Bank downloads arrive wide, with one column per year. Pivot tables and Power BI prefer long. Power Query's **Unpivot** converts one into the other in a single step.

# One column, one data type

TEXT
**Tirana**
Branch, Product

NUMBER
**12 · 1200**
Units, Price

DATE
**03/10/2026**
Date of sale

TRUE / FALSE
**TRUE**
Paid?

### Excel decides cell by cell

One column can hide text among numbers. Clue: a number sitting on the left of its cell is stored as text.

### Power Query decides per column

It sets one type for the whole column and flags every value that doesn’t fit as an error.

02

# What makes data dirty

Missing values · duplicates · inconsistent categories · wrong types and formats · impossible values

PART 2 · DIRTY DATA

# When dirty data gets expensive

2013 · ECONOMICS

### A debt study’s missing countries

A famous paper linked high public debt to slow growth. A student replicating it found an average that skipped several countries. The headline result weakened.

2020 · PUBLIC HEALTH, UK

### Lost test results

Almost 16,000 positive COVID-19 tests were left out of England’s daily figures. The data passed through the old .xls format, which stops at 65,536 rows.

2012 · BANKING

### The “London Whale”

A JPMorgan risk model ran on spreadsheets fed by copy-paste, and a formula error understated the risk. Trading losses reached about $6 billion.

None of these errors was clever: a cell range, a file format, a copy-paste. Ordinary checks would have caught them.

# Six questions to ask of any dataset

### Accuracy

Are the values correct?

### Completeness

Is anything missing that should be there?

### Consistency

Is the same thing written the same way?

### Validity

Do values follow the rules? Units ≥ 0, dates are real dates.

### Uniqueness

Is each record there only once?

### Timeliness

Is it recent enough for the question?

PROBLEM 1 · COMPLETENESS

# Missing values

Missing data hides in many forms:

*blank* .. N/A - 0 9999

World Bank downloads mark missing values with “..”

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**Leave it empty** and report how many are missing.

**Remove the row** if only a few rows are missing, at random.

**Fill it** only with a rule you can state and defend.

Units: 10, 20, *blank*

AVERAGE = 15

Units: 10, 20, 0

AVERAGE = 10

Zero is a value. Blank means “we don’t know”.

PROBLEM 2 · UNIQUENESS

# Duplicates

| Branch | Product | Units | Price | Revenue |
|---|---|---|---|---|
| Tirana | Coffee beans | 12 | 1,200 | 14,400 |
| Durrës | Green tea | 5 | 650 | 3,250 |
| Tirana | Coffee beans | 12 | 1,200 | 14,400 |
| Vlorë | Green tea | 10 | 650 | 6,500 |

Total with the repeat

38,550 ALL

True total

24,150 ALL

### Exact duplicates

Every cell identical. Remove Duplicates handles them.

### Near duplicates

“tirana” vs “Tirana”, two date formats. Clean the text first, then remove duplicates.

### Not every repeat is an error

Two customers can buy the same thing on the same day. A transaction ID tells them apart.

PROBLEM 3 · CONSISTENCY

# Inconsistent spelling and categories

PIVOT · UNITS BY BRANCH · BEFORE

| Branch as typed | Units |
|---|---|
| Tirana | 12 |
| tirana | 5 |
| TIRANA | 3 |
| Tiranë | 7 |
| “Tirana ” (space at end) | 4 |

AFTER CLEANING

| Branch | Units |
|---|---|
| Tirana | 31 |

Five rows for one city. The real total, 31, appears nowhere in the “before” table.

**TRIM** removes extra spaces

**PROPER** fixes capitals

**Replace** or a lookup list: Tiranë → Tirana, ALB → Albania

PROBLEM 4 · VALIDITY

# Wrong types and formats

“1,200”

### Numbers stored as text

SUM skips them without warning, so totals come out too low.

1.234,5 · 1,234.5

### Decimal comma or point

Albanian settings use a comma for decimals, English files a point. The wrong setting changes the size of numbers.

03/10/2026

### Day or month first?

3 October or 10 March? Store real dates; how they look is a separate choice.

Also check **units**: thousands or millions? lek or euro? Put the unit in the column name, for example Price (ALL) or GDP (current US$).

PROBLEM 5 · ACCURACY

# Impossible and suspicious values

## Impossible

Units = −3

Date = 31/02/2026

Age = 250

Break a rule, so they must be wrong. Fix or remove them, and write down what you did.

## Suspicious

Green tea price = 65,000

(usually 650)

A typo, a unit mix-up, or a real event? Check the source before you change anything.

How to find them: Sort | MIN and MAX | Filter | Conditional formatting | Column profile in Power Query

Investigate before you delete. An outlier can be an error — or the most important row in the table.

EXERCISE

# Find the problems

Find at least six. For each one: what is wrong, and how would you fix it?

| | A · Date | B · Branch | C · Product | D · Units | E · Price (ALL) |
|---|---|---|---|---|---|
| 2 | 03/10/2026 | Tirana | Coffee beans | 12 | 1,200 |
| 3 | 2026-10-03 | tirana | coffee beans | 12 | 1200 |
| 4 | 04/10/2026 | Shkoder | Coffee Beans | | 1,200 |
| 5 | 04/10/2026 | Tiranë | Olive oil | -3 | 1,500 |
| 6 | 05/10/2026 | Vlorë | Green tea | 10 | 65000 |
| 7 | 05/10/2026 | Durres | Olive oil | six | 1,500 |
| 8 | 31/02/2026 | Vlorë | Green Tea | 4 | 650 |

03

# Cleaning with a method

A workflow you can repeat · quick fixes in Excel · Power Query · today’s lab

# A cleaning workflow in four steps

1

**Keep the raw file**

Save the original untouched. Work on a copy.

2

**Inspect**

Sort, filter, count blanks, check MIN and MAX. List the problems.

3

**Clean**

Fix one problem at a time: text first, then duplicates, then types.

4

**Check and document**

Compare row counts and totals before and after. Write down every step.

Same raw data + same steps = same clean table. That is what makes your work reproducible, and checkable by a lecturer, a manager or an auditor.

# Each problem has a tool

| Problem | Quick fix in Excel | Step in Power Query |
|---|---|---|
| Extra spaces | TRIM( ) | Transform › Format › Trim |
| Mixed capitals | PROPER( ), UPPER( ) | Transform › Format › Capitalize Each Word |
| Missing as “..” | Find and Replace | Replace Values: “..” → null |
| Blank rows | Filter, COUNTBLANK( ) | Home › Remove Rows › Remove Blank Rows |
| Duplicates | Data › Remove Duplicates | Home › Remove Rows › Remove Duplicates |
| Numbers as text | VALUE( ), Text to Columns | Change the column type to Decimal Number |

An Excel fix changes the cells once. A Power Query step is saved and runs again every time.

PART 3 · CLEANING WITH A METHOD

# Power Query: cleaning as a recipe




New data next month? Click **Refresh** and every step runs again: week 3's algorithm, saved.

LAB · POWER QUERY WITH WORLD BANK DATA

# Today’s lab, step by step

STEP 1

## Import

Data › Get Data › From File. Open the World Bank file and choose Transform Data.

STEP 2

## Headers

Use First Row as Headers, if Power Query hasn’t done it already.

STEP 3

## Missing values

Replace “..” with null, so Excel treats it as empty.

STEP 4

## Blanks and repeats

Remove Blank Rows, then Remove Duplicates.

STEP 5

## Data types

Text for country, whole number for year, decimal number for values.

STEP 6

## Load and check

Close and Load to a new sheet. Compare the row count with the raw file.

☆ **Extension:** Unpivot the year columns so each row is one country in one year.

# AI can help you clean. You still check.

## Ask AI to…

- Write a formula: “Excel formula to trim spaces and capitalise the names in column B”
- Explain a Power Query step you don’t understand
- Suggest which problems to look for in a column

## Then you…

- Compare row counts and totals before and after
- Never paste personal or confidential data into an AI tool
- Record the tool and what it did in your AI-use statement

TO REMEMBER

# Four things to take into the lab

1. One row = one observation, one column = one variable, one cell = one value.
2. Blank is not zero. Decide how to treat missing data, and say so.
3. Clean text before removing duplicates. Check totals after.
4. Never edit the raw file. Record your steps so they can run again.

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PROJECT

Cleaned dataset with documented Power Query steps, due week 7

READING

Microsoft Learn: Power Query, intro module

NEXT WEEK

Analysis and visualisation: pivot tables and charts